

Neural Particle Method: Experimental Record

Synthetic-data setup, machinery, and the benchmark suite

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Abstract

This note records, in one place, the synthetic markets and dynamics used throughout the project, the simulation, estimation and scoring machinery, and the specification of the clean benchmark suite agreed on 8 September 2026 (Section 4), run on 9 September 2026; its tables are generated from the MLflow store. The suite follows the discovery, in the diagnostics of 8 September, that all one-pass schemes were fitting the conditional expectation of the raw Euler variance rather than of its positive part, which the dynamics actually use (Section 3.4); the diagnostic runs themselves live in the MLflow experiments `mechanism`, `bump_floor` and `bump_floor_nn`.

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1 Conventions

Explicit denotes the one-pass, per-slice calibration with a per-slice neural network. Nadaraya–Watson (NW) is the classical particle method: the same one-pass structure with kernel regression. Other per-slice estimators are named by their estimator (RKHS, spline, ridge). *Implicit* denotes the damped outer iteration with a single network in (t, x) . All errors are implied-volatility root-mean-square errors in volatility basis points against the target surface, unless stated otherwise. “Pooled” means over all quoted strikes and maturities together; “wings” means the quoted strikes with $|k| > 0.25$; per-maturity columns restrict to one maturity.

2 Synthetic data

2.1 Target surfaces: SSVI power law

Each scenario’s market is an SSVI implied-variance surface in the power-law parameterisation. With log-moneyness $k = \log(K/S_0)$, $S_0 = 1$, and parameters $(\sigma_0, \eta, \gamma, \rho_s)$,

$$\theta_T = \sigma_0^2 T, \quad \varphi(\theta_T) = \eta \theta_T^{-\gamma}, \quad w(k, T) = \frac{1}{2} \theta_T \left(1 + \rho_s \varphi k + \sqrt{(\varphi k + \rho_s)^2 + 1 - \rho_s^2} \right), \quad (1)$$

where w is total implied variance and the implied volatility is $\sqrt{w(k, T)/T}$. Parameters are drawn independently per scenario from

$$\sigma_0 \sim U(0.12, 0.35), \quad \eta \sim U(0.5, 2.0), \quad \gamma \sim U(0.3, 0.5), \quad \rho_s \sim U(-0.85, -0.10),$$

resampled until the sufficient no-arbitrage conditions $0 < \gamma \leq \frac{1}{2}$, $|\rho_s| < 1$, $\eta(1 + |\rho_s|) \leq 2$ hold. Scenario s_i uses the seed $1000 + i$ (`bench/scenarios.py`, `make_registry`).

The Dupire local volatility is analytic from Equation (1): with w_k, w_{kk}, w_T the derivatives of w ,

$$\sigma_{\text{Dup}}^2(T, k) = \frac{w_T}{1 - \frac{k}{w} w_k + \frac{1}{4} \left(-\frac{1}{4} - \frac{1}{w} + \frac{k^2}{w^2} \right) w_k^2 + \frac{1}{2} w_{kk}}, \quad (2)$$

with the denominator floored at 0.05, the local variance clipped to $[10^{-8}, 9]$, and the time argument clipped to $[t_{\min}, T]$ with $t_{\min} = 0.004$ (`market/local_vol.py`). These clipping constants are part of the calibrated behaviour and are never changed.

2.2 Variance dynamics

The calibrated model is Heston-type LSV in log-spot $X = \log S$,

$$dX_t = -\frac{1}{2} L^2(t, X_t) V_t dt + L(t, X_t) \sqrt{V_t} dW_t, \quad dV_t = \kappa(\theta - V_t) dt + \xi \sqrt{V_t} dB_t, \quad d\langle W, B \rangle_t = \rho dt. \quad (3)$$

For scenario s_i the dynamics parameters are drawn from the same seed as the surface: $\kappa \in \{1, 2, 3\}$, $\xi \in \{0.3, 0.6, 1.0\}$, $\rho \in \{-0.7, -0.3\}$, each uniform over its set, and $\theta = v_0 = \sigma_0^2$ so that the variance level matches the surface’s short-end level. The Feller ratio $2\kappa\theta/\xi^2$ therefore ranges from 0.05 to 4.3 across the grid. Table 1 lists all twenty scenarios. Horizon $T = 2$ years; spot $S_0 = 1$; no rates or dividends.

Two further families exist in the registry and were not used today: six variants of s01 with (ξ, ρ) swept over the grid (`fig3_registry`), and three Heston-market scenarios whose target surface is generated by a Heston model with parameters different from the calibrated dynamics (`li_simple`, `li_complex`, `bayer`; horizon 1 year; Dupire surface from prices on a 50×121 grid).

2.3 Quotes, bump, and scoring

Quoted strikes are 13 log-strikes equally spaced in log between 0.6 and 1.6; maturities are 0.25, 0.5, 1 and 2 years. The wings are the strikes with $|k| > 0.25$ (six of thirteen). The warm-start

sid	SSVI surface				dynamics			Feller
	σ_0	η	γ	ρ_s	κ	ξ	ρ	
s01	0.261	0.52	0.34	-0.21	1	0.3	-0.7	1.51
s02	0.208	1.04	0.45	-0.56	3	0.6	-0.7	0.72
s03	0.163	0.89	0.40	-0.35	3	1.0	-0.7	0.16
s04	0.120	0.70	0.33	-0.56	2	0.6	-0.3	0.16
s05	0.285	1.01	0.49	-0.72	2	1.0	-0.3	0.32
s06	0.304	1.00	0.38	-0.58	3	1.0	-0.3	0.55
s07	0.202	0.56	0.46	-0.49	3	1.0	-0.3	0.25
s08	0.158	1.72	0.34	-0.11	1	1.0	-0.7	0.05
s09	0.133	0.99	0.31	-0.64	3	1.0	-0.3	0.11
s10	0.142	1.30	0.48	-0.11	3	1.0	-0.7	0.12
s11	0.144	1.10	0.38	-0.11	2	1.0	-0.7	0.08
s12	0.191	1.01	0.33	-0.22	3	1.0	-0.7	0.22
s13	0.303	0.62	0.43	-0.51	2	1.0	-0.3	0.37
s14	0.220	1.38	0.48	-0.11	1	0.6	-0.3	0.27
s15	0.127	0.72	0.37	-0.33	3	0.3	-0.3	1.08
s16	0.172	1.16	0.37	-0.42	1	0.6	-0.7	0.17
s17	0.136	1.13	0.42	-0.50	1	0.3	-0.3	0.41
s18	0.257	1.18	0.47	-0.24	3	0.6	-0.3	1.10
s19	0.170	1.09	0.39	-0.23	2	0.3	-0.7	1.28
s20	0.253	0.93	0.32	-0.32	3	0.3	-0.7	4.27

Table 1: The twenty synthetic scenarios: SSVI surface parameters, variance dynamics, and the Feller ratio $2\kappa\theta/\xi^2$.

experiments bump the surface to $\sigma'_0 = \sigma_0 + 0.01$, $\eta' = 0.95\eta$, $\rho'_s = \min(\rho_s + 0.03, -0.05)$, a level rise of one volatility point with curvature and skew nudged inside the no-arbitrage region.

Scoring reprices vanilla calls under the calibrated leverage field on fresh paths: 500 000 particles, 200 Euler steps over $[0, 2]$, seed 10 000 plus a strategy-specific offset, call price as the sample mean of the payoff, Black–Scholes inversion at the grid-snapped maturity, and the exact SSVI implied volatility at the same snapped time as the target (`pricing/reprice.py`, `pricing/metrics.py`). The bump experiments use 300 000 repricing paths. Errors are reported as RMSE in volatility bp, pooled, wings-only, and per maturity.

3 Machinery

3.1 Simulation

Calibration and repricing share one stepper. With $\Delta = T/K$, $v^+ = \max(v, 0)$, $z_1 = \rho z_b + \sqrt{1 - \rho^2} z_p$ and independent standard normals z_b, z_p ,

$$x_{k+1} = x_k - \frac{1}{2}L_k(x_k)^2 v_k^+ \Delta + L_k(x_k) \sqrt{v_k^+} \sqrt{\Delta} z_1, \quad v_{k+1} = v_k + \kappa(\theta - v_k^+) \Delta + \xi \sqrt{v_k^+} \sqrt{\Delta} z_b. \quad (4)$$

This is full truncation: the raw v may be negative and is carried, but only v^+ enters any coefficient. The paper calibrated with $K = 50$ steps; the benchmark suite of Section 4 uses 200 for calibration and repricing alike. The leverage is stored per slice on a grid in log-spot and evaluated by linear interpolation, constant beyond the grid ends; the slice in force at time t is the last grid time $\leq t$. The explicit and NW passes use a per-slice grid at the 0.1% to 99.9% quantiles of the current cloud (101 points); the implicit loop uses a fixed grid of 81 points from log 0.4 to log 2.2. The leverage is capped at $L_{\max} = 4$ and the fitted conditional expectation is floored at 10^{-4} .

3.2 One-pass schemes and their estimators

At slice k , fit $f_k(x) \approx \mathbb{E}[V_{t_k} \mid X_{t_k} = x]$ on a subsample of 30 000 particles from the current cloud (the earlier default of the code; the benchmark suite of Section 4 fits on the full cloud), set $L_k = \text{clip}(\sigma_{\text{Dup}}(t_k, \cdot) / \sqrt{\max(f_k, 10^{-4})}, 0, L_{\text{max}})$, and advance the cloud. Slice 0 uses $f_0 \equiv v_0$. Estimators:

- **NW**: Gaussian Nadaraya–Watson with Silverman bandwidth $h = 1.06 \hat{\sigma}_x n^{-1/5}$ on the subsample.
- **Explicit (per-slice network)**: a $1 \rightarrow 64 \rightarrow 64 \rightarrow 1$ SiLU network with softplus output scaled by 0.04, input scaled by 0.3, trained by Adam at learning rate 10^{-2} on the unweighted squared loss; 400 steps on the first fitted slice and 120 steps on each later slice, warm-started from the previous slice’s weights.
- **RKHS**: kernel ridge with a Gaussian kernel of variance 0.1 in log-spot, 100 centres at the $j \cdot 100/101$ percentiles of the cloud, regularisation $\lambda = 10^{-9}$ (Bayer et al.’s published choices).
- **Ridge head on a frozen body**: the implicit network’s body (below) as a 65-dimensional feature map, with a per-slice ridge readout, $\lambda = 10^{-3}$ per sample, shrunk toward the previous slice’s coefficients.

3.3 Implicit loop

A single network $f_\theta(t, x)$, $2 \rightarrow 64 \rightarrow 64 \rightarrow 1$ SiLU with softplus output scaled by 0.04, inputs $(t/T, x/0.3)$. Starting from the explicit output as L^0 , each outer iteration simulates a full cloud under L^n with fresh random numbers, draws a pooled subsample of 60 000 points across all slices, trains θ for 300 Adam steps at learning rate 10^{-2} with the network and optimiser state carried over from the previous iteration, reads $\Phi(L^n)_k = \text{clip}(\sigma_{\text{Dup}}(t_k, \cdot) / \sqrt{\max(f_\theta(t_k, \cdot), 10^{-4})}, 0, L_{\text{max}})$ on the fixed grid, and sets $L^{n+1} = (1 - \alpha)L^n + \alpha\Phi(L^n)$ with $\alpha = \frac{1}{2}$. The paper uses six iterations. The pooled data are truncated, v^+ , before training.

3.4 The variance-target discrepancy

Before 8 September 2026 the one-pass schemes handed the estimator the raw Euler v , while the implicit loop handed it v^+ . Since only v^+ drives the dynamics Equation (4), the quantity the leverage must divide by is $\mathbb{E}[V^+ \mid X]$. Under Feller violation a sizeable fraction of particles has $v < 0$ after a coarse step (on s11, $v_0 = 0.02$, $\xi = 1$, $\Delta = 0.04$: the one-step standard deviation of v is 0.028), so $\mathbb{E}[v \mid X] < \mathbb{E}[v^+ \mid X]$ by the mean negative excursion, the denominator was too small, and the leverage too large. The effect is independent of the particle count and shrinks with the step size. Fitting on v^+ (`ExplicitConfig.fit_v_floor`) has been the package default since 8 September 2026; every result in Sections 4 and 5 uses it, and the archived replays of the paper’s runs pin it off explicitly so that they still reproduce bit for bit.

4 Benchmark suite

This section specifies the clean benchmark agreed on 8 September 2026 and run on 9 September 2026 with the variance floor of Section 3.4 as the package default. Every cell of the tables below is one MLflow run, keyed by method, scenario, seed and (for the lagged suite) lag type and offline particle count, in experiments `suite_cold`, `suite_offline`, `suite_lagged` and `pde_reference`. Every trained object is stored as a run artifact (network weights as a `torch` state dictionary, per-slice estimators and leverage fields as pickles and JSON) and can be reloaded by run id. The tables are generated from the store by `nparticle suite tables`.

4.1 Common setup

- **Scenarios.** The 20 SSVI scenarios of Table 1 and the three Heston-market scenarios used in the literature comparisons (`li_simple`, `li_complex`, `bayer`; a Heston market surface, a different Heston variance process as the calibrated model, one-year maturity, log-strikes on $[-0.9, 0.9]$). The Heston block is reported separately because it scores a single one-year maturity (the same 13 quoted strikes).
- **Time grid.** 200 Euler steps for calibration and 200 for repricing, $\Delta = 0.01$ on $[0, 2]$, so that every method sits at its step-count plateau (the step-count diagnostics of 8 September 2026, MLflow experiment `mechanism`) and the comparison is about the estimator alone.
- **Particles.** 100 000 particles for every cold row and for every online stage; every per-slice fit and every online head is fitted on the full cloud; the earlier code default of a 30 000-particle fit subsample is not used. Offline bodies at 200 000 and 500 000 particles, fitted on 100 000-particle subsamples.
- **Seeds.** Two seeds (0, 1) for every cold row and every online stage; one seed for each offline body. The reprice seed is 10 000 plus the calibration seed.
- **Floor.** All one-pass schemes fit $\mathbb{E}[V^+ | X]$; `fit_v_floor` becomes the default in this work and the tiny golden replays are regenerated in the same commit.
- **Implicit configuration.** Warm start from the explicit network field, 30 damped iterations at $\alpha = \frac{1}{2}$ with fresh noise per iteration, persistent network and optimiser (the best practical setting found in the convergence diagnostics of 8 September 2026, experiment `mechanism`).
- **Scoring.** As in Section 2.3: 500 000 repricing paths, Black–Scholes inversion at the snapped maturity, exact SSVI (or semi-analytic Heston) implied volatility as the target. Metrics per run: pooled and wings MAE and RMSE in volatility bp, both per maturity, the count of failed inversions, and the leverage RMSE against the PDE reference of the scenario. Errors are reported in implied-vol basis points (the headline) and, in the companion tables, as absolute option-price errors in basis points of spot, recovered exactly from the implied-vol errors through the Black–Scholes formula.
- **Latency.** Wall-clock seconds of the calibration only (repricing excluded). For the lagged suite, only the online stage is timed.
- **PDE reference.** The Fokker–Planck reference leverage (`reference/pde.py`) is solved per scenario at 200 steps and enters twice: repriced under the same Monte Carlo as every other row, which is the attainable floor of the scoring under this repricer (Gyöngy), and as the reference for the leverage-error column. Its repricing is repeated under a second reprice seed to measure the Monte Carlo noise floor of the score itself.

4.2 Cold suite

One calibration from scratch per method, scenario and seed at 100 000 particles. Methods, with their estimator of $\mathbb{E}[V^+ | X]$ per slice:

1. **NW:** Gaussian Nadaraya–Watson, Silverman bandwidth (Section 3.2).
2. **Explicit NN:** per-slice network, warm-started across slices (Section 3.2).
3. **Implicit NN:** damped global-network loop (Section 3.3, configuration above).
4. **RKHS:** Gaussian kernel ridge with 100 quantile centres, Bayer et al.’s published knobs (Section 3.2).
5. **Spline:** penalised cubic B-spline, 25 quantile knots, $\lambda = 1$.
6. **Literature block.** GHL kernel: Nadaraya–Watson in spot with the quartic kernel and the

Guyon–Henry–Labordère rule-of-thumb bandwidth $h(t) = 1.5 S_0 \sigma_{LV}(S_0, t) \sqrt{\max(t, 0.25)} N^{-1/5}$. Bins: piecewise-constant regression on 20 equal-frequency buckets (van der Stoep et al.). Muguruza: the kernel-free conditional Monte Carlo of Corollary 4.1, each particle contributing its one-step Gaussian conditional law of log-spot given the variance path. PU-RBF: Hakala’s partition-of-unity radial basis functions, 40 centres, 5-nearest-neighbour widths, regulariser 0.2.

7. **PDE**: the reference leverage repriced by Monte Carlo (attainable floor).

Table 2: Cold suite, SSVI scenarios: mean and median over the 20 scenarios and two seeds. The PDE row is the attainable floor of the scoring; its seed-to-seed spread (median over scenarios of the absolute difference in pooled MAE between the two reprice seeds) is 6.6 bp, the Monte Carlo noise floor of the score.

method	pooled MAE (bp)		pooled RMSE (bp)		wings MAE (bp)		lev. RMSE	latency (s)
	mean	median	mean	median	mean	median	median	median
NW	46	33	87	67	70	48	0.093	8.0
Explicit NN	57	36	112	64	86	49	0.089	1042.7
Implicit NN	58	37	110	73	88	47	0.395	1298.6
RKHS	81	70	131	110	113	99	0.224	12.4
Spline	47	34	90	64	73	46	0.098	2.7
GHL kernel	45	32	88	62	70	45	0.106	8.5
Bins	66	61	120	99	105	84	0.210	1.5
Muguruza	154	139	188	171	147	136	0.161	10.6
PU-RBF	57	40	112	66	90	53	0.130	2.5
PDE (attainable floor)	42	27	85	59	62	35	0.000	0.0

Table 3: Cold suite, SSVI scenarios, the same runs in price space: absolute option-price error in basis points of spot (spot is one), mean over scenarios and seeds. Relative price error is not shown; in the far wings the target prices are of order 10^{-6} of spot.

method	pooled mean	pooled median	near-money	wings	$T = 2$	worst quote
NW	2.7	2.4	3.6	1.8	4.3	9.3
Explicit NN	2.6	2.1	3.4	2.0	3.7	8.6
Implicit NN	3.3	2.7	4.6	2.1	5.5	12.2
RKHS	8.5	8.1	11.6	5.7	19.0	32.3
Spline	2.5	2.1	3.3	1.7	3.8	8.2
GHL kernel	2.5	2.2	3.4	1.7	4.0	8.6
Bins	2.7	2.3	3.3	2.2	4.2	8.5
Muguruza	29.1	35.0	47.1	12.3	62.7	122.9
PU-RBF	2.8	2.5	3.4	2.2	4.4	9.2
PDE (attainable floor)	2.2	1.7	3.2	1.3	3.1	8.1

4.3 What a lag is

An overnight calibration is done on surface S_0 ; the intraday market is S_1 , and the question is how much of the overnight work survives. Because the synthetic surfaces are parametric, a lag is a move in parameter space and never a bump of quoted points, so arbitrage cannot be introduced by accident. Two lags are run for every cell.

Surface lag. For SSVI, the existing bump: $\sigma'_0 = \sigma_0 + 0.01$, $\eta' = 0.95\eta$, $\rho'_s = \min(\rho_s + 0.03, -0.05)$, shrunk by quarters until the Gatheral–Jacquier sufficient conditions ($\eta(1 + |\rho_s|) \leq 2$, $0 < \gamma \leq \frac{1}{2}$) hold; calendar no-arbitrage is automatic since $\theta_T = \sigma_0^2 T$ is increasing. For a Heston market, the matching move in vol terms: $\sqrt{v_0}$ and $\sqrt{\theta}$ up one volatility point, $\rho \rightarrow \min(\rho + 0.03, -0.05)$, $\xi \rightarrow 0.95\xi$; the surface is model-generated and arbitrage-free for any valid parameters.

Surface plus spot lag. The surface lag together with a sticky-strike spot move of +2%, $S_1 = S_0 e^\delta$, $\delta = \log 1.02$. Under sticky-strike the implied volatility at each absolute strike is unchanged, so in log-moneyness relative to the new spot the surface is translated: $\sigma'_{IV}(k, T) = \sigma_{IV}(k + \delta, T)$. A translation

Table 4: Cold suite, Heston-market scenarios (one-year maturity, same 13 quotes): mean and median over the three scenarios and two seeds..

method	pooled MAE (bp)		pooled RMSE (bp)		wings MAE (bp)		lev. RMSE	latency (s)
	mean	median	mean	median	mean	median	median	median
NW	193	61	218	66	125	64	0.145	7.9
Explicit NN	189	55	214	60	120	57	0.143	1044.4
Implicit NN	179	51	206	56	107	51	0.528	1281.5
RKHS	209	67	242	74	147	77	0.241	12.2
Spline	193	57	218	62	124	60	0.165	2.7
GHL kernel	194	56	218	60	126	58	0.145	7.8
Bins	186	61	218	73	117	71	0.345	1.5
Muguruza	381	460	408	494	293	398	0.753	8.2
PU-RBF	193	55	218	60	124	57	0.145	2.3
PDE (attainable floor)	189	40	211	44	123	43	0.000	0.0

Table 5: Cold suite, Heston-market scenarios, the same runs in price space: absolute option-price error in basis points of spot (spot is one), mean over scenarios and seeds. Relative price error is not shown; in the far wings the target prices are of order 10^{-6} of spot.

method	pooled mean	pooled median	near-money	wings	worst quote
NW	50.5	13.2	89.3	17.2	135.1
Explicit NN	50.0	12.2	88.4	17.1	134.2
Implicit NN	48.4	11.7	86.4	15.9	131.8
RKHS	52.4	13.0	91.5	18.9	140.5
Spline	50.7	12.4	89.5	17.4	135.6
GHL kernel	50.6	12.4	89.4	17.4	135.5
Bins	48.6	11.4	86.9	15.8	132.3
Muguruza	93.0	110.5	157.1	38.0	223.2
PU-RBF	50.8	12.4	89.6	17.5	135.7
PDE (attainable floor)	48.9	8.5	86.7	16.5	130.6

in k preserves Durrleman’s butterfly condition (it involves only w and its k -derivatives at a point) and calendar monotonicity, so the shifted surface is arbitrage-free. Call prices at fixed strike change with the spot, so the Dupire local volatility is that of the translated surface, $\tilde{w}(k) = w(k + \delta)$ with k measured from the new forward; the particles also start at the new spot. The stored denominators $f(t, x) = \mathbb{E}[V_t | X_t = x]$ were fitted for a cloud started at S_0 , and a conditional expectation given spot depends on the starting point: with $\rho < 0$ a particle sitting 2% above the old start had its variance drift down, so the stale f at the new spot is below v_0 , the stale leverage is too high near the money, most at short maturities. Sticky-delta, by contrast, leaves the surface in moneyness unchanged and the whole calibration is scale invariant, so a sticky-delta spot move is invisible to every method; the surface-only lag already covers it. A 2% daily move is one to one-and-a-half standard deviations at 20–30% volatility.

4.4 Lagged suite

Offline stage (experiment `suite_offline`, one run per scenario, body type and particle count): a full calibration on S_0 at 200 000 or 500 000 particles, 200 steps, seed 0: either the explicit NN pass (the per-slice networks and the per-slice denominators f_k are saved) or the implicit loop (the global network is saved). These runs are cached and reused by every online cell below.

Online stage (experiment `suite_lagged`, one run per scenario, method, offline size, lag and online seed): given the intraday surface S_1 and 100 000 fresh particles, build the intraday leverage and reprice. Only this stage is timed. Methods:

1. **Explicit NN, stale f , fresh Dupire:** $L_k = \text{clip}(\sigma_{\text{Dup}}^{S_1}(t_k, \cdot) / \sqrt{\max(f_k, 10^{-4})}, 0, L_{\max})$ with the offline per-slice f_k . Costs nothing online beyond the repricing.

2. **Implicit NN, stale f , fresh Dupire**: the same with $f = f_\theta(t, \cdot)$ from the offline global network.
3. **Explicit NN with RKHS head**: simulate the 100 000 online particles under the stale- f leverage of item 1 with fresh Dupire, slice by slice; at each slice fit a Gaussian kernel ridge (kernel width 0.5 times the cloud’s standard deviation, ridge 10^{-3}) to the residual $v^+ - f_k(x)$ on the online cloud, and use $\max(f_k + \text{residual}, \frac{1}{4}f_k)$ as the denominator for the next step; the floor keeps the head from driving the leverage to its cap on Feller-violating scenarios.
4. **Implicit NN with RKHS head**: the same residual head on the global network’s f_θ .
5. **Explicit NN with ridge head**: the per-slice network body frozen as a feature map (64 features plus a constant) with its last layer refitted by ridge on the online cloud, shrunk toward the offline readout. The body’s output is $\text{softplus}(Aw) \cdot 0.04$, and the refit minimises $\sum_i (\text{softplus}(A_i w) \cdot 0.04 - v_i^+)^2 + \lambda_{\text{eff}} \|w - w_0\|^2$ on the raw variance scale by Gauss–Newton to convergence (at most 20 steps) from the offline last layer w_0 , with $\lambda_{\text{eff}} = \lambda$ times the mean diagonal of the Gauss–Newton curvature at w_0 and $\lambda = 10^{-3}$, so the penalty is a fixed fraction of the data curvature whatever the output scale and cloud size. The denominator is $\text{softplus}(Aw) \cdot 0.04$ at the converged w . When the online data agree with the offline fit the first step is zero and $w = w_0$.
6. **Implicit NN with ridge head**: the same on the global body in (t, x) .
7. **Explicit NN with spline head and Implicit NN with spline head**: the same residual correction with the penalised cubic B-spline of Section 3.2 (25 quantile knots, $\lambda = 1$) in place of the kernel ridge, with the same floor.

The online reference rows are NW re-solved from scratch on S_1 at 100 000 particles and 200 steps (the best intraday strategy in the floor-corrected bump runs, experiment `bump_floor`), and the stale leverage with no action.

4.5 Per-scenario numbers

Cold suite, SSVI scenarios, pooled MAE (bp)

sid	NW	Explicit NN	Implicit NN	RKHS	Spline	GHL kernel	Bins	Muguruza	PU-RBF
s01	15	13	16	17	14	14	23	13	14
s02	12	15	20	38	14	14	20	30	27
s03	61	107	58	160	65	63	101	238	115
s04	106	104	149	120	111	105	144	187	107
s05	50	43	39	59	45	44	62	137	44
s06	26	24	24	104	27	25	34	82	26
s07	47	45	49	69	51	46	79	185	47
s08	75	124	81	130	78	77	112	435	110
s09	136	127	172	214	136	136	160	260	138
s10	29	36	80	94	30	29	61	344	36
s11	71	204	145	160	80	72	127	333	134
s12	53	52	73	83	57	51	85	229	86
s13	27	26	22	60	28	27	36	137	27
s14	24	26	32	32	24	25	21	140	26
s15	71	71	69	71	71	72	77	72	72
s16	36	37	35	109	39	34	69	163	62
s17	26	26	28	37	27	26	43	38	25
s18	18	19	30	32	18	19	15	28	19
s19	22	22	24	21	23	22	35	23	24
s20	11	9	11	11	11	10	19	10	9

Table 6: Lagged suite, SSVI scenarios: mean over the 20 scenarios and two online seeds; latency is the online stage only. One block per lag type..

method	offline N	pooled MAE	pooled RMSE	wings MAE	$T = 0.25$ MAE	online latency (s)
<i>Surface lag</i>						
Stale leverage, no action	–	99	133	105	141	0.0
NW re-solve on S_1	–	44	82	64	96	8.5
Explicit NN, stale f , fresh Dupire	200k	55	101	78	121	0.0
Explicit NN, stale f , fresh Dupire	500k	55	102	77	123	0.0
Implicit NN, stale f , fresh Dupire	200k	62	113	91	128	0.0
Implicit NN, stale f , fresh Dupire	500k	67	114	94	137	0.0
Explicit NN + RKHS head	200k	46	90	69	108	17.0
Explicit NN + RKHS head	500k	46	91	69	109	17.1
Implicit NN + RKHS head	200k	47	93	72	108	17.1
Implicit NN + RKHS head	500k	44	85	65	102	17.0
Explicit NN + ridge head	200k	47	93	71	113	30.1
Explicit NN + ridge head	500k	48	95	72	117	28.4
Implicit NN + ridge head	200k	50	98	75	118	33.1
Implicit NN + ridge head	500k	44	85	65	103	32.6
Explicit NN + spline head	200k	43	83	63	99	7.5
Explicit NN + spline head	500k	43	82	63	98	7.3
Implicit NN + spline head	200k	46	89	70	103	7.2
Implicit NN + spline head	500k	42	81	62	96	7.3
<i>Surface plus 2% sticky-strike spot lag</i>						
Stale leverage, no action	–	97	137	108	165	0.0
NW re-solve on S_1	–	44	83	65	98	8.5
Explicit NN, stale f , fresh Dupire	200k	97	139	114	198	0.0
Explicit NN, stale f , fresh Dupire	500k	97	141	115	200	0.0
Implicit NN, stale f , fresh Dupire	200k	98	143	121	192	0.0
Implicit NN, stale f , fresh Dupire	500k	97	140	119	184	0.0
Explicit NN + RKHS head	200k	46	90	70	110	16.9
Explicit NN + RKHS head	500k	46	92	70	111	17.1
Implicit NN + RKHS head	200k	49	94	74	110	17.1
Implicit NN + RKHS head	500k	45	87	68	101	17.0
Explicit NN + ridge head	200k	46	90	68	109	31.5
Explicit NN + ridge head	500k	48	93	71	114	29.3
Implicit NN + ridge head	200k	49	94	72	110	34.1
Implicit NN + ridge head	500k	46	90	68	109	32.8
Explicit NN + spline head	200k	44	84	65	100	7.5
Explicit NN + spline head	500k	44	84	65	100	7.3
Implicit NN + spline head	200k	49	92	75	104	7.3
Implicit NN + spline head	500k	56	100	87	108	7.3

Cold suite, Heston scenarios, pooled MAE (bp)

sid	NW	Explicit NN	Implicit NN	RKHS	Spline	GHL kernel	Bins	Muguruza	PU-RBF
bayer	507	502	478	521	511	515	481	661	513
li_complex	61	55	51	67	57	56	61	460	55
li_simple	12	10	8	39	12	11	15	22	11

Lagged suite, surface lag, SSVI scenarios, pooled MAE (bp)

[illegible]

Lagged suite, surface plus 2% spot lag, SSVI scenarios, pooled MAE (bp)

5.2 Training the body

Setup. The offline bodies of Section 4.4 at 500 000 particles were retrained on the six scenarios under two training budgets and refitted on the full online cloud: “short” is the paper’s schedule (400 Adam steps on the first slice, 120 on each later slice; 300 steps per outer iteration for the implicit loop), “long” is 2000 and 1000 steps per slice (1000 per iteration on a 200 000-point pool for the implicit loop). Table 11 scores each body as a cold calibrator on its own overnight surface; Table 12 runs the lagged suite on it with every fit on the full 100 000-particle online cloud, alongside the first pass’s 30k-subsample rows.

Conclusions. (i) Training the explicit body ten times longer turns it from the weakest cold calibrator into the best one on these scenarios (64 $\rightarrow$ 50 pooled against NW’s 53), at 2.5 hours per body with full-batch training; the minibatch schedule of Section 5.1 should reach the same quality in a fraction of that and is untested as a body. (ii) The implicit loop gets worse when trained harder (66 $\rightarrow$ 88) and its bodies are unstable across retrains; it is not a usable offline source. (iii) On the long-trained body the heads sit on or just under the NW re-solve under both lags, the RKHS head best; the stale body alone remains useless after the spot move. (iv) The RKHS head of the first pass had no positivity guard and a fixed kernel width, and drove the denominator to its floor on the Feller-violating scenarios (s09: 858 bp on the long-trained body); the guard and the adaptive width of Section 4.4 were introduced here, and the suite’s RKHS rows were recomputed with them.

5.3 The online budget

Setup. The long-trained explicit body at 500 000 particles with each head, against NW re-solved from scratch, at online budgets of 10 000, 30 000 and 80 000 particles (every fit on the whole cloud), both lags, six scenarios, two seeds; the 100 000 column is Table 12. Experiment `suite_budget`.

Conclusions. At 100 000 particles every method sits within a few basis points of the floor and the comparison cannot separate them. As the online cloud thins, NW degrades (49 $\rightarrow$ 57 bp from 100k to 10k) and the body with a head does not (47 $\rightarrow$ 50): at 10 000 particles the RKHS head is 7 bp ahead under both lags, 9 bp in the wings and 6 bp at two years, and the body with either kernel head at 10k matches NW at 80k, with eight times fewer particles and a seventh of the online time. The head is doing the work, not the body: the stale body alone is flat at 61 bp and useless after the spot move at every budget. The ridge head is the weakest at 10k, where it refits 65 features on the thinnest data.

5.4 What the studies say

With the variance floor in place, the classical one-pass kernels are the best cold calibrators and sit a few basis points above the scheme’s own floor, so at 100 000 particles no estimator can look better than “a few basis points”. A per-slice network is only competitive once it is trained properly, and then it is the best cold calibrator. A frozen, properly trained body with an online head reprices today’s vanillas at least as well as a fresh NW calibration at any online budget, and clearly better when the online budget is small, which is where the latency argument for the head lives. The implicit loop is not a useful offline source. The remaining distance to a 10 bp tolerance on liquid quotes is the scheme’s, not the estimator’s.

Table 7: Lagged suite, SSVI scenarios, the same runs in price space: absolute option-price error in basis points of spot (spot is one), mean over scenarios and seeds. Relative price error is not shown; in the far wings the target prices are of order 10^{-6} of spot.

method	offline N	pooled	near-money	wings	worst quote
<i>Surface lag</i>					
Stale leverage, no action	—	16.8	26.9	7.7	61.0
NW re-solve on S_1	—	3.0	4.4	1.9	11.0
Explicit NN, stale f , fresh Dupire	200k	4.9	7.2	2.8	16.0
Explicit NN, stale f , fresh Dupire	500k	4.9	7.2	2.8	16.1
Implicit NN, stale f , fresh Dupire	200k	4.9	7.3	2.8	17.5
Implicit NN, stale f , fresh Dupire	500k	5.8	8.8	3.2	20.6
Explicit NN + RKHS head	200k	2.8	3.9	1.8	9.6
Explicit NN + RKHS head	500k	2.8	3.9	1.8	9.7
Implicit NN + RKHS head	200k	2.8	3.9	1.8	9.7
Implicit NN + RKHS head	500k	2.8	3.9	1.7	9.7
Explicit NN + ridge head	200k	2.7	3.6	1.8	9.4
Explicit NN + ridge head	500k	2.6	3.5	1.8	9.1
Implicit NN + ridge head	200k	2.9	4.0	1.9	10.1
Implicit NN + ridge head	500k	2.8	3.9	1.8	10.2
Explicit NN + spline head	200k	2.7	3.9	1.7	9.7
Explicit NN + spline head	500k	2.7	3.9	1.7	9.7
Implicit NN + spline head	200k	2.8	3.9	1.8	9.7
Implicit NN + spline head	500k	2.7	3.9	1.7	9.7
<i>Surface plus 2% sticky-strike spot lag</i>					
Stale leverage, no action	—	13.7	21.5	6.5	52.4
NW re-solve on S_1	—	3.1	4.5	1.9	11.2
Explicit NN, stale f , fresh Dupire	200k	10.5	16.6	5.2	30.7
Explicit NN, stale f , fresh Dupire	500k	10.5	16.6	5.2	30.5
Implicit NN, stale f , fresh Dupire	200k	10.1	15.8	5.0	29.9
Implicit NN, stale f , fresh Dupire	500k	10.7	16.9	5.4	33.1
Explicit NN + RKHS head	200k	2.8	3.9	1.8	9.7
Explicit NN + RKHS head	500k	2.8	3.9	1.8	9.7
Implicit NN + RKHS head	200k	2.8	3.9	1.9	9.8
Implicit NN + RKHS head	500k	2.9	3.9	1.9	10.7
Explicit NN + ridge head	200k	2.9	4.1	1.8	10.9
Explicit NN + ridge head	500k	2.9	4.2	1.8	11.1
Implicit NN + ridge head	200k	3.1	4.4	1.9	11.6
Implicit NN + ridge head	500k	3.0	4.2	1.9	11.3
Explicit NN + spline head	200k	2.8	3.9	1.8	9.8
Explicit NN + spline head	500k	2.8	3.9	1.8	9.8
Implicit NN + spline head	200k	2.9	3.9	2.0	10.4
Implicit NN + spline head	500k	3.1	4.1	2.2	11.4

Table 8: Lagged suite, Heston-market scenarios..

method	offline N	pooled MAE	pooled RMSE	wings MAE	online latency (s)
<i>Surface lag</i>					
Stale leverage, no action	–	297	323	210	0.0
NW re-solve on S_1	–	233	258	160	8.6
Explicit NN, stale f , fresh Dupire	200k	250	273	174	0.0
Explicit NN, stale f , fresh Dupire	500k	247	270	170	0.0
Implicit NN, stale f , fresh Dupire	200k	233	260	155	0.0
Implicit NN, stale f , fresh Dupire	500k	236	262	158	0.0
Explicit NN + RKHS head	200k	234	258	161	17.1
Explicit NN + RKHS head	500k	234	258	161	17.1
Implicit NN + RKHS head	200k	232	257	158	17.2
Implicit NN + RKHS head	500k	232	257	157	17.1
Explicit NN + ridge head	200k	232	256	160	28.7
Explicit NN + ridge head	500k	232	256	159	27.7
Implicit NN + ridge head	200k	229	254	154	36.4
Implicit NN + ridge head	500k	227	252	155	33.9
Explicit NN + spline head	200k	234	258	161	7.7
Explicit NN + spline head	500k	234	258	161	7.7
Implicit NN + spline head	200k	233	258	160	7.3
Implicit NN + spline head	500k	232	257	159	7.4
<i>Surface plus 2% sticky-strike spot lag</i>					
Stale leverage, no action	–	281	313	195	0.0
NW re-solve on S_1	–	225	248	154	8.5
Explicit NN, stale f , fresh Dupire	200k	227	253	155	0.0
Explicit NN, stale f , fresh Dupire	500k	225	251	152	0.0
Implicit NN, stale f , fresh Dupire	200k	217	245	140	0.0
Implicit NN, stale f , fresh Dupire	500k	214	241	138	0.0
Explicit NN + RKHS head	200k	225	248	154	17.2
Explicit NN + RKHS head	500k	225	248	154	17.3
Implicit NN + RKHS head	200k	224	247	151	17.3
Implicit NN + RKHS head	500k	224	247	150	17.2
Explicit NN + ridge head	200k	221	244	150	30.7
Explicit NN + ridge head	500k	221	244	150	30.4
Implicit NN + ridge head	200k	218	243	145	38.8
Implicit NN + ridge head	500k	218	242	146	39.0
Explicit NN + spline head	200k	226	249	155	7.6
Explicit NN + spline head	500k	226	249	155	7.6
Implicit NN + spline head	200k	226	250	154	7.3
Implicit NN + spline head	500k	225	248	153	7.3

Table 9: Lagged suite, Heston-market scenarios, the same runs in price space: absolute option-price error in basis points of spot (spot is one), mean over scenarios and seeds. Relative price error is not shown; in the far wings the target prices are of order 10^{-6} of spot.

method	offline N	pooled	near-money	wings	worst quote
<i>Surface lag</i>					
Stale leverage, no action	—	75.9	132.4	27.5	186.9
NW re-solve on S_1	—	60.3	105.3	21.8	155.0
Explicit NN, stale f , fresh Dupire	200k	64.3	111.8	23.5	160.8
Explicit NN, stale f , fresh Dupire	500k	63.7	111.2	23.1	160.1
Implicit NN, stale f , fresh Dupire	200k	61.5	107.7	22.0	157.6
Implicit NN, stale f , fresh Dupire	500k	62.1	108.3	22.5	157.5
Explicit NN + RKHS head	200k	60.6	105.5	22.0	155.2
Explicit NN + RKHS head	500k	60.5	105.5	22.0	155.1
Implicit NN + RKHS head	200k	60.5	105.4	21.9	155.0
Implicit NN + RKHS head	500k	60.5	105.5	21.9	155.2
Explicit NN + ridge head	200k	60.1	104.6	22.0	154.6
Explicit NN + ridge head	500k	60.0	104.5	21.9	154.3
Implicit NN + ridge head	200k	60.2	104.9	21.9	155.1
Implicit NN + ridge head	500k	59.6	103.5	22.0	153.7
Explicit NN + spline head	200k	60.5	105.4	22.0	155.2
Explicit NN + spline head	500k	60.5	105.4	22.0	155.2
Implicit NN + spline head	200k	60.5	105.4	22.0	155.2
Implicit NN + spline head	500k	60.5	105.4	22.0	155.2
<i>Surface plus 2% sticky-strike spot lag</i>					
Stale leverage, no action	—	70.9	126.0	23.6	179.8
NW re-solve on S_1	—	58.5	102.0	21.1	152.2
Explicit NN, stale f , fresh Dupire	200k	58.1	103.2	19.4	148.7
Explicit NN, stale f , fresh Dupire	500k	57.7	102.6	19.1	147.8
Implicit NN, stale f , fresh Dupire	200k	57.3	102.3	18.8	147.8
Implicit NN, stale f , fresh Dupire	500k	56.8	101.1	18.8	145.7
Explicit NN + RKHS head	200k	58.6	102.3	21.2	152.4
Explicit NN + RKHS head	500k	58.6	102.2	21.2	152.3
Implicit NN + RKHS head	200k	58.7	102.4	21.2	152.5
Implicit NN + RKHS head	500k	58.7	102.4	21.2	152.7
Explicit NN + ridge head	200k	57.6	100.5	20.9	150.4
Explicit NN + ridge head	500k	57.6	100.3	20.9	150.3
Implicit NN + ridge head	200k	58.1	101.3	21.0	152.0
Implicit NN + ridge head	500k	57.7	100.4	21.1	150.9
Explicit NN + spline head	200k	58.8	102.4	21.4	152.8
Explicit NN + spline head	500k	58.8	102.5	21.5	152.8
Implicit NN + spline head	200k	58.9	102.4	21.5	152.8
Implicit NN + spline head	500k	58.8	102.4	21.4	152.8

Table 10: Relative error of the fitted denominator against the exact conditional expectation, in percent, and seconds per fit.

estimator	s02 wings		s11 wings		s11 centre	
	$t = 0.25$	$t = 1$	$t = 0.25$	$t = 1$	mean	s / fit
NW, 30k particles	3.9	6.0	17.5	10.3	14.0	0
NW, 100k	2.8	3.5	14.6	9.5	15.0	0
NW, 500k	1.9	1.6	14.9	6.1	13.7	1
Spline, 100k	4.6	4.1	15.4	12.6	13.1	0
RKHS, 100k	4.1	9.6	29.8	12.7	34.7	0
Net 64×2 , 120 full-batch steps (paper)	6.2	2.6	55.1	16.2	33.8	6
Net 64×2 , 1000 full-batch steps	2.2	1.3	54.6	10.0	28.7	52
Net 64×2 , 20000 minibatch steps	0.6	1.7	12.7	9.0	14.9	145
Net 64×3 , 120 full-batch steps	1.9	1.6	55.3	22.5	30.0	9
Net 64×3 , 20000 minibatch steps	0.7	1.8	10.4	9.7	15.2	230
Net 256×2 , 20000 minibatch steps	1.0	1.5	8.0	10.5	15.3	520

Table 11: Offline bodies at 500 000 particles as cold calibrators, six scenarios, seed 0: pooled and wings MAE (bp), the two-year slice, and training time.

training	body	pooled MAE	wings MAE	$T = 2$ MAE	train s
Short training, 30k fits	explicit	64	93	17	346
Short training, 30k fits	implicit	66	96	28	756
Short training, 100k fits	explicit	70	102	18	1262
Short training, 100k fits	implicit	78	115	21	1712
Long training, 100k fits	explicit	50	72	18	9115
Long training, 100k fits	implicit	88	131	26	11671

Table 12: Lagged suite on the 500 000-particle explicit body, six scenarios, two online seeds, pooled MAE (bp), by body training and fit sample. Heads are the guarded versions of Section 4.4.

online method	surface lag			surface plus spot lag		
	short, 30k fits	short, 100k fits	long, 100k fits	short, 30k fits	short, 100k fits	long, 100k fits
NW re-solve	52	49	49	52	48	48
Explicit body, stale f , fresh Dupire	64	65	61	119	118	99
Explicit body + RKHS head	53	56	47	53	55	45
Explicit body + spline head	52	48	48	51	48	49
Explicit body + ridge head	63	59	48	57	60	47

Table 13: Pooled MAE (bp) by online particle budget, and the online time at 10 000 particles.

online method	surface lag, pooled MAE				surface plus spot lag				s at 10k
	10k	30k	80k	100k	10k	30k	80k	100k	
NW re-solve	57	52	50	49	56	51	49	48	1.0
Explicit body, stale f , fresh Dupire	61	61	61	61	99	99	99	99	0.0
Explicit body + spline head	51	50	48	48	51	49	49	49	0.9
Explicit body + RKHS head	50	48	47	47	49	48	47	45	2.1
Explicit body + ridge head	54	48	49	48	56	49	48	47	2.9